

PROGRAM ASSESSMENT REPORT

v2024.1

Tracking data

CIM code	
Program name	Master of Data Science
year	AY 24/25
Program assessment coordinator	Yuhan Ding
Assessment plan year	AY 24/25
No assessment this year per plan	☐
LO(s) assessed	For LO 3 and LO 4, see the description below.
Presented at the program authority meeting date: <i>Please attach minutes</i>	

As outlined in the AY22-23 assessment document, we will evaluate the following program goals for AY 24-25:

LO3: Students will be able to analyze data via the appropriate methods and make conclusions based on the analysis.

LO4: Students will be able to modify existing machine learning or statistical models to meet customers' needs.

According to the assessment plan, CSP 572 Data Science Practicum (Spring 25) will assess Learning Outcomes 3 and 4.

Methodology

For **LO 3**, students will be able to analyze data using the appropriate methods and draw conclusions based on the analysis.

Learning Objective 3 was evaluated through **CSP 572: Data Science Practicum**, a project-based course offered during the spring 2025 semester. A total of 40 students, all enrolled in the Master of Data Science program, were divided into ten groups.

These groups were evaluated based on their final project presentations, which took place during the last week of the course. The assessment criteria included the selection of methods, analysis and results conducted by the students.

- Rubrics

Criteria

3 – Exemplary

2 – Proficient

1 – Developing

Selection of Methods	Selects the most appropriate methods for the data and question, clearly justifying the choice.	Uses methods generally appropriate, with minor mismatches or limited justification.	Uses inappropriate or unclear methods; justification is missing or incorrect.
Application of Analysis	Performs analysis accurately; correctly implements procedures and checks necessary assumptions.	Performs analysis mostly correctly, with minor errors, and demonstrates a basic understanding of key steps.	The analysis contains multiple errors or misunderstandings and lacks evidence of correct application.
Interpretation of Results	Interprets results accurately and insightfully, linking findings clearly to the research question.	Interprets results correctly but with limited depth or incomplete connection to the research question.	Misinterprets results or fails to connect them meaningfully to the research question.
Conclusions and Implications	Draws well-supported conclusions; acknowledges limitations and suggests practical implications or future directions.	Draws reasonable conclusions that generally follow from analysis; may overlook limitations or broader implications.	Conclusions are unsupported, unclear, or unrelated to the analysis.

Using the rubric above, we apply the following criteria to determine how well the learning objective has been achieved.

Group	Point Range	Description
High Exemplary /	10–12	Strong performance across almost all criteria; analysis is accurate, methods are appropriate, and conclusions are clear and insightful.
Medium Proficient /	7–9	Adequate performance; minor errors or gaps in method, application, interpretation, or conclusions.

Low / Developing 4–6

The performance reveals significant gaps; the methods or analysis are flawed, the interpretations are weak, and the conclusions are unsupported.

Results

The table below displays the number of groups that received each category in the final presentation.

Rubric Category	Number of groups
Low / Developing	0
Medium / Proficient	7
High / Exemplary	3

The table below displays the number and percentage of students at each proficiency level.

Rubric Category	Assessments Categories	Number of groups	Percentage of students in each category.
Low/ Developing	Does not meet expectations	0	0
Medium/ Proficient	Meets expectations	7	70%
High/ Exemplary	Proficient	3	30%

The data shows that all students meet the learning objective. These results indicate that students have satisfactorily developed the ability to choose appropriate methods for analyzing data and drawing conclusions based on the analysis. We have included two group artifacts for "Meets Expectations" and one group artifact for "Proficient."

- Not Meet Expectation
N.A.
- Meet Expectation (Artifact 1)

Group 03: AI-Powered Product Recommendation System

Project Descriptions:

- **Description:** Build a recommendation engine that suggests products to customers based on their browsing and purchase history.

- **Tasks:** Build recommendation algorithms, work with product metadata, and create a recommendation dashboard for stakeholders.

Rubric Category: Medium/ Proficient

Selection of Methods	Application of Analysis	Interpretation of Results	Conclusions and Implications	Final Point
3	2	2	2	8

- Meet Expectation (Artifact 2)

Group 04: Personalized Cart Item Recommendations

Project Descriptions: Develop a recommendation system that suggests additional products to add to a customer's cart based on their current cart items, browsing history, and previous purchases. The goal is to increase cart value by recommending relevant cross-sell or upsell products.

Tasks:

- Analyze the relationship between products frequently purchased together.
- Suggest products to customers based on their current cart items and the most frequently bought items.

Rubric Category: Medium/ Proficient

Selection of Methods	Application of Analysis	Interpretation of Results	Conclusions and Implications	Final Point
2	2	2	2	8

- Proficient (Artifact 3)

Group 07: Account Imputation and Clustering

Project Descriptions: We have numerous contact and account attributes that require imputation. Contact attributes include job functions and management level that are derived from titles provided. Account attributes include industries, sub-industries, NAICS and SIC codes etc. that should where missing fields should be imputed. Once you have the missing attributes imputed you should use these to perform clustering to come up with segments which will be used to hyper-personalize our email marketing campaigns. What we currently have is such clustering, but it doesn't include imputed contact and account attributes.

INSIGHTS FROM CLUSTERING

- **Cluster 0:**
 - Low sales amount (\$849), few orders (2.51), minimal emails and event interactions.
 - Average time spent is 68.6 seconds.
 - **Interpretation:** This cluster represents low-engagement users/customers who may need more nurturing or targeting.
- **Cluster 1:**
 - Moderate sales amount (\$655,134), significant orders (8790.75), but no email or event engagement.
 - **Interpretation:** These are high-value customers with transactional behavior, but no interest in other communication channels (e.g., email, events).
- **Cluster 2:**
 - Zero sales and orders, heavy webpage engagement (15,545 interactions), and the highest time spent (576,530 seconds).
 - **Interpretation:** These may be leads exploring content but not converting into buyers.
- **Cluster 3:**
 - Highest sales amount (\$1.85M) and order count (277), but no engagement in other categories.
 - **Interpretation:** This cluster represents top-performing customers - high spenders with focused activity.

Data provided includes sales transactions data, email marketing tables, website traffic and web order mediums.

Rubric Category: High/ Exemplary

Selection of Methods	Application of Analysis	Interpretation of Results	Conclusions and Implications	Final Point
3	3	3	3	12

Continuous Improvement, CI, plans

The assessment data highlights essential insights into our students' performance on specific learning objectives. Here's an analysis of the findings and our improvement plans:

Learning Objective 3: 100% of student teams have successfully met the learning objective, indicating that all groups are competent in selecting appropriate analytical methods and drawing valid conclusions from their data analyses. This result demonstrates strong collaborative skills and effective application of analytical techniques in a project-based context. Moving forward, we aim to maintain this high level of achievement while promoting more advanced reasoning, methodological creativity, and critical reflection within team settings.

Plan for Improvement:

Although all teams performed at a high level, continuous growth can be achieved by emphasizing deeper understanding, analytical sophistication, and innovation. Planned improvements include:

1. **Expanding Data Complexity:** Introduce more open-ended, less structured datasets that require teams to make strategic, collaborative analytical decisions.
2. **Strengthening Method Justification:** Require teams to provide written or oral justifications for their choice of analytical approaches, discussing alternatives and limitations.
3. **Integrating Advanced Methods:** Encourage exploration of advanced statistical or computational techniques (e.g., model comparison, feature selection, or sensitivity analysis).
4. **Enhancing Reflective Practice:** Include short team reflections on collaboration dynamics, analytical reasoning, and lessons learned from challenges encountered.
5. **Connecting to Real-World Contexts:** Partner with external stakeholders or use case studies to reinforce the importance of aligning analysis and conclusions with practical decision-making needs.

Methodology

For **LO 4**, students will be able to modify existing machine learning or statistical models to meet customers' needs.

Learning Objective 3 was evaluated through **CSP 572: Data Science Practicum**, a project-based course offered during the spring 2025 semester. A total of 40 students, all enrolled in the Master of Data Science program, were divided into ten groups.

These groups were evaluated based on their final project presentations, which took place during the last week of the course. The assessment criteria included understanding the requirements, model selection, implementation, evaluation, and validation.

- Rubrics

Criteria	3 – Exemplary	2 – Proficient	1 – Developing
Understanding Customer Requirements	Clearly identifies customer needs; considers constraints and objectives; aligns model modifications with goals.	Identifies most customer needs but may miss some details or constraints; partially aligns model modifications to goals.	Misunderstands or overlooks customer needs; model modifications are misaligned with goals.

Model Selection and Modification	Appropriately selects the base model and implements modifications correctly to address the customer's requirements.	Selects and modifies the model with minor errors or a partial understanding of the impact.	Chooses inappropriate models or misapplies modifications, lacking an understanding of their implications.
Technical Implementation	Implements modifications efficiently, correctly, and reproducibly; code or workflow is clear and well-documented.	Implements modifications with minor errors; code or workflow is mostly transparent and reproducible.	Implementation is error-prone, unclear, or incomplete, making it difficult to reproduce results.
Evaluation and Validation	Thoroughly evaluates the modified model against customer needs, interprets the results correctly, and ensures that performance meets the requirements.	Evaluates the modified model using basic validation; the results align with customer needs.	Fails to adequately evaluate or validate the model; performance may not meet customer needs.

Using the rubric above, we apply the following criteria to determine how well the learning objective has been achieved.

Group	Point Range	Description
High / Exemplary	10–12	Strong performance across all criteria; modifications fully address customer needs, are technically correct, evaluated thoroughly, and well-documented.
Medium / Proficient	7–9	Adequate performance; minor errors or gaps in understanding customer needs, model modification, implementation, or evaluation.
Low / Developing	4–6	Performance shows significant gaps; modifications are misaligned with customer needs, technical implementation is flawed, and evaluation is inadequate.

Results

The table below displays the number of groups that received each category in the final presentation.

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Low / Developing	0
Medium / Proficient	7
High / Exemplary	3

The table below displays the number and percentage of students at each proficiency level.

Rubric Category	Assessments Categories	Number of groups	Percentage of students in each category.
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High/ Exemplary	Proficient	3	30%

The data shows that all students meet the learning objective. These results indicate that students have satisfactorily developed the ability to choose appropriate methods for analyzing data and drawing conclusions based on the analysis. We have included two group artifacts for "Meets Expectations" and one group artifact for "Proficient."

- Not Meet Expectation

N.A.

- Meet Expectation (Artifact 4)

Group 02: AI-Powered Product Recommendation System

Project Descriptions:

CNT works on enabling housing and commercial developments near transit. We were instrumental in popularizing the idea of doing this work with a focus on equity and thinking about factors beyond location. To continue this work, we are seeking support in developing a machine learning framework for identifying which areas of the city are most suitable for equitable transit-oriented development projects. We currently have a tool, the Equitable Transit Oriented Development Social Impact Calculator, that provides an overlay of relevant data sources, and we would like to take it further by adding vehicle miles traveled data and assessed property value data. Additional data to consider includes:

- Aerial Imagery
- LIDAR\Landcover
- Traffic Speed (Replica or IDOT)
- Cook County Assessor Property Valuation data
- City of Chicago Zoning

Rubric Category: Medium/ Proficient

Understanding Customer Requirements	Model Selection and Modification	Technical Implementation	Evaluation and Validation	Final Point
3	3	2	1	9

- Meet Expectation (Artifact 5)

Group 06: AI-Powered Bibu Chatbot

Project Descriptions: Develop an AI-driven chatbot that provides customer support, answers product-related queries, and assists with order issues in real time. The bot would support multiple languages, providing seamless service to customers from different regions.

Tasks:

- Build a conversational AI model using NLP to understand customer queries and provide accurate responses.
- Integrate with your platform to provide real-time order tracking, returns, and shipping updates.
- Continuously improve the chatbot's capabilities using reinforcement learning and customer feedback.

Rubric Category: Medium/ Proficient

Understanding Customer Requirements	Model Selection and Modification	Technical Implementation	Evaluation and Validation	Final Point
3	2	3	3	11

- Proficient (Artifact 6)

Group 01: Data Science for the game of soccer/football

Project Descriptions: The project will take spatiotemporal tracking data from soccer/football matches and synthesize it into metrics that can be used to make an expect goals (xG) framework. Students will work in multiple phases:

1. Data Engineering
2. Data Visualization
3. Feature Engineering
4. Model Building

Rubric Category: High/ Exemplary

Understanding Customer Requirements	Model Selection and Modification	Technical Implementation	Evaluation and Validation	Final Point
3	2	3	3	11

Continuous Improvement, CI, plans

The assessment data highlights essential insights into our students' performance on specific learning objectives. Here's an analysis of the findings and our improvement plans:

Learning Objective 4: 100% of student teams successfully met the learning objective, demonstrating strong ability to modify existing machine learning or statistical models to meet customer needs. This indicates that all teams are proficient in understanding stakeholder requirements, selecting appropriate models, and implementing effective modifications. The results reflect high levels of technical competence, teamwork, and practical application of analytical skills in real-world problem-solving contexts. Moving forward, we seek to sustain this excellent performance while advancing students' capabilities in model innovation, evaluation, and communication with non-technical audiences.

Plan for Improvement:

Although all teams achieved the objective, continuous improvement can be pursued by deepening students' understanding of model design, optimization, and professional communication. Planned enhancements include:

1. **Incorporating More Complex Scenarios:** Use datasets and customer needs that require balancing multiple objectives (e.g., accuracy, interpretability, and computational efficiency).
2. **Encouraging Model Innovation:** Challenge students not only to modify existing models but also to propose creative enhancements or hybrid approaches.

3. **Emphasizing Model Evaluation:** Require teams to conduct thorough performance assessments, including sensitivity analysis and validation under different conditions.
4. **Strengthening Communication Skills:** Include deliverables that involve clearly and persuasively explaining model decisions and results to non-technical stakeholders.
5. **Integrating Ethical and Responsible AI Considerations:** Introduce reflection on bias, fairness, and transparency in model modification and deployment.

Overall Summary and Continuous Improvement Plan

Both assessed learning objectives — *“Analyze data via the appropriate methods and make conclusions based on the analysis”* and *“Modify existing machine learning or statistical models to meet customers’ needs”* — achieved a 100% success rate. These results demonstrate that all student teams are performing at a high level of competency in data analysis, model adaptation, and evidence-based decision-making. The outcomes reflect strong technical proficiency, effective teamwork, and the ability to apply analytical reasoning to real-world challenges.

To ensure continued growth and sustained excellence, future improvement efforts will focus on expanding analytical depth, promoting model innovation, and strengthening professional communication. Planned actions include introducing more complex and ambiguous datasets, requiring teams to justify methodological choices, encouraging creative model enhancements, and incorporating reflective and ethical components into project work. These steps aim to deepen understanding, foster critical and innovative thinking, and prepare students for increasingly sophisticated analytical tasks in professional settings.